

A REPORT ON STATISTICAL CHALLENGES IN ONLINE EXPERIMENTS

INTRODUCTION

In response to the growing demand of online businesses to engage in large scale data-driven decision making, Nicholas Larsen, Jonathan Stallrich, Srijan Sengupta, Alex Deng, Ron Kohavi, and Nathaniel T. Stevens review the open challenges and methodologies in their article, Statistical Challenges in Online Controlled Experiments: A Review of A/B Testing Methodology. They highlight several open challenges in the field with the aim of drawing greater attention from statisticians. The primary goal of the article is to increase collaboration between academia and the tech industry. This report will first summarize some of the authors' key arguments and then present a detailed exposition of the Sequential Probability Ratio Test (SPRT), comparing it to the standard Likelihood Ratio Test (LRT).

SUMMARY OF THE ARTICLE

The authors begin by defining terms and outlining key statistics that shape the current technological landscape. Online Controlled Experiments (OCEs), which they define as digital versions of randomized controlled trials, are the engines driving innovation in internet-based products and services. Major tech companies run these experiments at large scales to inform decisions that maximize revenue and optimize the user experience. The article attributes the rise in popularity of OCEs to their relatively low cost compared to traditional, physical experimentation. In fact, the authors argue that for organizations with expertise in software development, the marginal cost of running experiments may be negligible, allowing them to conduct hundreds of tests concurrently. Because of the inevitable growth of the importance of internet based products and hence the OCEs, it is crucial for academic researchers to understand these processes and contribute to solving the associated statistical challenges.

At the core of online controlled experiments lies the comparison of two variants: a treatment and a control. Units in the experiments such as users or web sessions are randomly assigned to either variant, and a response variable is measured for each. In most cases the metric measured is an average and the experiments are used to estimate average treatment effect (ATE). The authors define ATE as the difference in average outcome when the treatment is applied globally and when the control is applied globally. Formally, the ATE is given by:

$$\begin{aligned}\tau &= \mu(1) - \mu(0) \\ &= \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)),\end{aligned}$$

where:

$Y_i(0)$: potential response when unit i receives the control,

$Y_i(1)$: potential response when unit i receives the treatment,

$\mu(0)$: expected population outcome under control,

$\mu(1)$: expected population outcome under treatment.

The analysis of A/B tests typically relies on two-sample hypothesis testing for τ . A key assumption throughout the process is the Stable Unit Treatment Value Assumption (SUTVA), which asserts that a user's response is independent of the treatment assignments of other users. In practice, however, SUTVA is not always satisfied, and this violation is one of several challenges the authors highlight.

The first challenge discussed is sensitivity and small treatment effects. Contrary to popular belief, the authors argue that online controlled experiments suffer from inadequate sample sizes and as a result, even major companies are sometimes unable to design adequately powered experiments. To address low statistical power, research focuses on improving sensitivity either by reducing the variance of Y_i or by lowering the variance of estimators for τ . One approach presented in the article involves transforming Y into a new variable Y^* with lower variance. Another technique involves defining a more efficient estimator of τ using the method of control variates. In the context of OCEs, this method is known as Controlled experiments Utilizing Pre-Experiment Data (CUPED). The authors also discuss triggered analysis, whereby only users who could potentially be impacted by an experiment are considered in the analysis. However, using triggered analysis presents problems such as generalizing the result to a broader population and reducing the variance of τ estimators to offset the smaller sample sizes that result from triggering.

The next challenge is Heterogeneous Treatment Effects (HTEs), which occur when treatment effects on subgroups differ from the population level Average Treatment Effect. Several challenges arise in estimating HTEs: testing for HTEs tends to risk inflated Type I error rates; when users are not randomized into subgroups, distinguishing correlation from causation becomes difficult; and small treatment effects often make online studies under-powered, resulting in high false negative rates for subgroup effects. In addition to the Stable Unit Treatment Value Assumption (SUTVA), analyses involving HTEs typically assume unconfoundedness, which the article defines as: $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp W_i \mid X_i$, meaning that the response is independent of the treatment assignment W_i conditional on the covariate. The authors note that HTE estimation is a promising area for machine learning and mention double machine learning as one of the techniques used to estimate it.

Long-term effects is presented as another open challenge. In most online controlled experiments, short-term treatment effects observed during or immediately after deployment may not necessarily be the same as the long-term effects. To help build on this idea, the authors define novelty effect and primacy effect. A novelty effect exists when a novel change is initially intriguing, leading to increasing engagement but diminishes over time whereas, a primacy effect exists when the initial reaction to a change is not positive but over time users get used to the change and engagement increases. One proposed solution to this problem is to identify and track short-term driver metrics that are causally linked to long-term outcomes.

The next open problem is interference, which occurs when treatments *intefere* with each other causing SUTVA to be violated. The article identifies two types of interference: network interference, which arises when units are connected through a social network, and marketplace interference, which occurs when units compete for shared resources in two-sided markets. To address interference, the authors discuss using experimental designs that reduce the amount of interference and analysis methods that provide unbiased treatment effect estimates in the presence of interference. Cluster-based randomization and switchback experiments are presented as some processes that achieve this.

Finally, the authors address optional stopping, a challenge especially relevant to real-time experiments. In online controlled experiments, estimates, confidence intervals, and p-values are often updated continuously as data streams in. In such settings, practices like “peeking” which is stopping the experiment as soon as a significant p-value is observed, can severely inflate false positive rates. This issue motivates the use of sequential testing methods. The article offers an overview of the classic Sequential Probability Ratio Test (SPRT) and the Mixture Sequential Probability Ratio Test (mSPRT). While the authors do not focus on it, they acknowledge that group sequential methods are valuable for improving decision making speed and controlling false positives in OCEs.

In the next section of this report, I will provide a more detailed exposition of the classic SPRT.

CLASSIC SEQUENTIAL PROBABILITY TEST

The Sequential Probability Ratio Test (SPRT) is a hypothesis testing method that allows decisions to be made by analyzing data sequentially, rather than fixing a sample size in advance (Chou 2024). This approach is particularly well-suited for online controlled experiments, where data arrives in real time and early decisions may be beneficial.

The basic steps of SPRT are as follows:

1. Define the hypotheses: null hypothesis (H_0), alternative hypothesis (H_A) and two constants A and B such that $0 < B < A$, where: $B = \frac{\beta}{1-\alpha}$ and $A = \frac{1-\beta}{\alpha}$.
2. Observe each data point at a time. After each observation, calculate the likelihood ratio test statistic at time n :

$$\Lambda_n = \prod_{i=1}^n \frac{f(y_i | \theta_1)}{f(y_i | \theta_0)},$$

where y_i are observed data from i.i.d. random variables $Y_i \sim f(y_i | \theta)$.

3. Check the Λ_n and make decision as follows:

- if $\Lambda_n > A$ reject H_0 and stop the test,
- if $\Lambda_n < B$ fail to reject H_0 and stop the test,
- if $B < \Lambda_n < A$ continue collecting and evaluating new data until the evidence is strong enough to reject or not reject H_0 .

With SPRT, the amount of data points required to come to a conclusion can be defined by a random variable. The boundary of the decision region depends on the expected value of this random variable, called the Average Sample Number (ASN). The ASN for the SPRT is lower than all other sequential tests and is usually lower than traditional, fixed-size sampling methods (Statistics How To. 2025).

The key advantages of the SPRT are that it allows making a decision as soon as sufficient evidence accumulates and is particularly useful when sampling is expensive or time-consuming. The main disadvantage of SPRT is that it is optimal only for simple hypotheses testing.

The example below is drawn from a Medium article by Carey Chou (2024), which provides a simplified walkthrough of the Sequential Probability Ratio Test.

Example 1 Suppose we are evaluating whether a new checkout flow improves the conversion rate on a website selling event tickets. The control group uses the current flow, while the test group uses the new one. A conversion is defined as a visitor completing a purchase.

Assume the baseline conversion rate under the null hypothesis is $p_0 = 0.10$, and the minimum practical effect size is a 20% increase, making the alternative conversion rate $p_1 = 0.12$. We set the significance level to $\alpha = 0.05$ and the desired power to $1 - \beta = 0.80$.

From this, we calculate the decision boundaries as:

$$A = \frac{1-\beta}{\alpha} = \frac{0.80}{0.05} = 16, \quad B = \frac{\beta}{1-\alpha} = \frac{0.20}{0.95} \approx 0.21$$

At each step, the likelihood ratio is computed as:

$$\Lambda_n = \frac{p_1^x(1-p_1)^{n-x}}{p_0^x(1-p_0)^{n-x}}$$

where x is the number of conversions, and n is the total number of visitors observed.

Sequence of Events:

- First visitor in both groups converts: $\Lambda_1 = \frac{0.12}{0.10} = 1.20$
- Second visitor in both groups does not convert: $\Lambda_2 = 1.20 \times \frac{0.88}{0.90} \approx 1.173$
- Third visitor does not convert: $\Lambda_3 = 1.173 \times \frac{0.88}{0.90} \approx 1.147$
- Fourth visitor converts: $\Lambda_4 = 1.147 \times \frac{0.12}{0.10} \approx 1.376$

The SPRT continues updating the likelihood ratio with each new observation. If it exceeds $A = 16$, the test stops and the null hypothesis is rejected. If it drops below $B \approx 0.21$, we fail to reject the null. Otherwise, we continue evaluating new visitors.

Brief recap of Likelihood Ratio Test

The following excerpt, sourced from Chapter 10 lecture notes, offers a general overview of the Likelihood Ratio Test (LRT).

Define a test statistic, λ , as:

$$\lambda = \frac{L(\hat{\Omega}_0)}{L(\hat{\Omega})}$$

A Likelihood Ratio Test (LRT) of $H_0 : \theta \in \Omega_0$ vs. $H_A : \theta \in \Omega_A$ employs λ as a test statistic. The rejection region (RR) is determined by:

$$\lambda \leq k,$$

where k is chosen appropriately.

If λ is close to 0, it indicates the likelihood is smaller under H_0 , suggesting we should reject H_0 .

An example drawn from Homework 6 is shown below to illustrate LRT:

Example 2 A survey of voter sentiment was conducted in four mid-city political wards to compare the fraction of voters favoring candidate A. Random samples of 200 voters were polled in each of the four wards, with the results as shown in the accompanying table. The numbers of voters favoring A in the four samples can be regarded as four independent binomial random variables. Construct a likelihood ratio test of the hypothesis that the fractions of voters favoring candidate A are the same in all four wards. Use $\alpha = 0.05$.

Ward	1	2	3	4	Total
Favor A	76	53	59	48	236
Do not favor A	124	147	141	152	564
Total	200	200	200	200	800

$$\alpha = 0.05, Y_i \sim \text{Binomial}(n_i, p_i) \text{ where } i \in [1, 4]$$

$\searrow$ $\perp$ RVs

$$n_1 = n_2 = n_3 = n_4 = 400$$

$H_0: p_1 = p_2 = p_3 = p_4 = p_c$ H_A : At least one differs

$$\hat{p}_c = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2 + n_3 \hat{p}_3 + n_4 \hat{p}_4}{n_1 + n_2 + n_3 + n_4} = \sum_{i=1}^4 \frac{y_i}{800}$$

$$L(p) = \prod_{i=1}^4 \binom{200}{y_i} p^{y_i} (1-p)^{200-y_i}$$

$$\lambda = \frac{L(\hat{p}_c)}{L(\hat{p}_0)} = \frac{\left(\frac{\sum y_i}{800}\right)^{\sum y_i} \left(1 - \frac{\sum y_i}{800}\right)^{800 - \sum y_i}}{\prod_{i=1}^4 \left(\frac{y_i}{200}\right) \left(1 - \frac{y_i}{200}\right)^{200 - y_i}}$$

Reject if $-2 \ln \lambda > k$, chosen for $\alpha = 0.05$ from $\chi^2(3-0) = \chi^2(3) \Rightarrow \chi^2_{0.05}(3) = 7.81$

Substituting $y_1 = 76, y_2 = 53, y_3 = 59, y_4 = 48$;
 $-2 \ln \lambda = 10.54$

$10.54 > 7.81$, therefore we reject H_0 .

We have sufficient evidence to conclude the proportion of voters favoring A is not the same in all 4 wards.

Figure 1: Solution of Example 2

COMPARISON OF SPRT AND LRT

The Sequential Probability Ratio Test (SPRT) and the Likelihood Ratio Test (LRT) are both rooted in the likelihood principle, but they differ significantly in implementation as can be seen in the image below (Khude 2019):

Recall that the standard LRT has critical region of the form

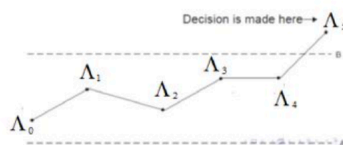
$$\Lambda_n = \lambda(X_1, X_2, \dots, X_n) = \log \frac{L(\theta_1; X_1, X_2, \dots, X_n)}{L(\theta_0; X_1, X_2, \dots, X_n)} > K$$

Wald's Sequential Probability Ratio Test (SPRT) has the following form:

If $\Lambda_n > B$ decide that H_A is true and stop;

If $\Lambda_n < A$ decide that H_0 is true and stop;

If $A \leq \Lambda_n \leq B$ collect another observation to obtain Λ_{n+1}



The key differences between the two processes are summarized in the table below:

SPRT (Sequential Probability Ratio Test)	Standard LRT (Likelihood Ratio Test)
Evaluates data as it arrives	Pre-specified fixed sample size
Can stop early based on A and B	Stops only after collecting the full sample
Continuously updated likelihood ratio: $\Lambda_n = \prod_{i=1}^n \frac{f(y_i \theta_1)}{f(y_i \theta_0)}$	Single likelihood ratio: $\lambda = \frac{L(\hat{\Omega}_0)}{L(\hat{\Omega})}$
Preferable in real-time and online experiments	Less suited for real-time environments
Online A/B testing, dynamic trials, real-time systems	Traditional hypothesis testing, clinical trials, offline analysis

CONCLUSION

This report reviewed key statistical challenges in online controlled experiments as outlined in the article, and presented a comparison between the Sequential Probability Ratio Test and the standard Likelihood Ratio Test. As the use of internet-based services and products continues to rise, increased collaboration between industry and academia will be even more critical.

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